# Coursework 1 – A Bookstore Inventory & Sales Manager

## Introduction

This is the first COMP1038 Coursework. It is worth **30% of the module mark**. It requires you to write an interactive, menu-driven program that manages a small bookstore's book information and sales records. The deadline for this exercise is **18:00 on Monday 24th of November 2025**.

**Read the entire document before beginning the exercise.**

If you have any questions about this exercise, please ask in the Q&A forum on Moodle, after a tutorial session, or during the advertised office hours. Do not post your program or parts of your program to Moodle as you are not allowed to share your coursework programs with other students. If any questions require this exercise to be clarified, then this document will be updated, and everyone will be notified via Moodle.

## Submission

You must submit a single C source code file containing all your code for this exercise. This file must be called **bookstore.c** and must not require any other files outside of the standard C headers. The first line of the file should be a comment which contains your student ID number, username, and full name, of the form:

```
// 6512345 zy12345 Joe Blogs
```

The file must compile without warnings or errors when I use the command

```
gcc -Wall -lm bookstore.c -o bookstore
```

This command will be run on X2Go. If it does not compile, for any reason, then you will lose all the marks for testing (common reasons in the past have been submitting a file with the wrong filename or developing your solution on your personal computer without having tested it on X2Go). If the file compiles but has warnings, then you will lose some marks for not correcting the warnings.

Before you submit your source code file, you must complete the **Coursework submission coversheet** on Moodle.

The completed source code file should be uploaded to the Coursework 1 Submission link on the COMP1038 Moodle page. You may submit as many times as you wish before the deadline (the last submission before the deadline will be used). After the deadline has passed, if you have already submitted your exercise, then you will not be able to submit again. If you have not already submitted, then you will be allowed to submit **once**.

**Late submissions**: Late submissions will lose 5 percentage points **per hour**, rounded up to the next whole hour. This is to better represent the large benefit a small amount of extra time can give at the end of a programming exercise. No late submissions will be accepted more than 24 hours after the exercise deadline. If you have extenuating circumstances, you should file them before the deadline.

## Plagiarism

You should complete this coursework on your own. Anyone suspected of plagiarism will be investigated and punished in accordance with the university policy on plagiarism (see your student handbook and the University Quality Manual). This may include a mark of zero for this coursework.

You should write the source code required for this assignment yourself. If you use code from other sources (books, web pages, etc), you should use comments to acknowledge this (and marks will be heavily adjusted down accordingly). *The only exception to this is the example programs given in lectures and tutorials; you may use them, with or without modification, without penalty.*

You must not copy or share source code with other students. You must not work together on your solution. You can informally talk about higher-level ideas but not to a level of detail that would allow you all to create the same source code.

Remember, it is quite easy for experienced lecturers to spot plagiarism in source code. If you are having problems, you should ask questions rather than plagiarize. If you are not able to complete the exercise, then you should still submit your incomplete program as that will still get you some of the marks for the parts you have done (but make sure your incomplete solution compiles and partially runs!).

## Task

Your task is to develop a bookstore management program to manage a small bookstore's book information and sales records. An executable file with the name 'bookstore' is available on Moodle. You should download it to your X2Go account and then run the commands below in the terminal.

```
chmod u+x bookstore

./bookstore
```

Your program should replicate all functionalities and produce identical output to the provided executable file for any user input. You are suggested to use the data structure as follows in your program:

```
#define MAX_BOOKS 200    // maximum number of books in store
typedef struct {
    int     id;          // unique identifier
    char    title[100];
    char    author[50];
    int     stock;       // inventory
    double  price;       // unit price
    int     sold;        // quantity sold (tracked during runtime)
} Book;
```

Do not use linked lists and file I/O in your program.

## Marking

The marking scheme will be as follows:

- *Tests (90%):* Your program should correctly implement all task requirements. A number of tests will be run against your program with different input data to verify that your program produces correct outputs in the correct format. The specific test cases will remain confidential, but general examples may include:
    - Does the program work with typical valid input?
    - Does the program correctly deal with invalid input values?
    - Does the program work with a proper format of output, i.e. exactly the same as the given executable file?

As noted in the submission section, if your program does not compile, **then you will lose all testing marks**. If the format of outputs is different from the given file, then you will lose up

to **50% of testing marks.** Note that even minor formatting errors in your program's output may result in a significant loss of marks.

- *Source code formatting (10%):* Your program should be correctly formatted and easy to understand by a competent C programmer. This includes, but is not limited to, indentation, bracketing, variable/function naming, and use of comments. See the textbook for examples of correctly formatting programs.

**END**